

## Space Physics Laboratory Logbook

### Princeton University - 171 Broadmead

**April 23, 2026 (Thursday)**

07:58 – LYK

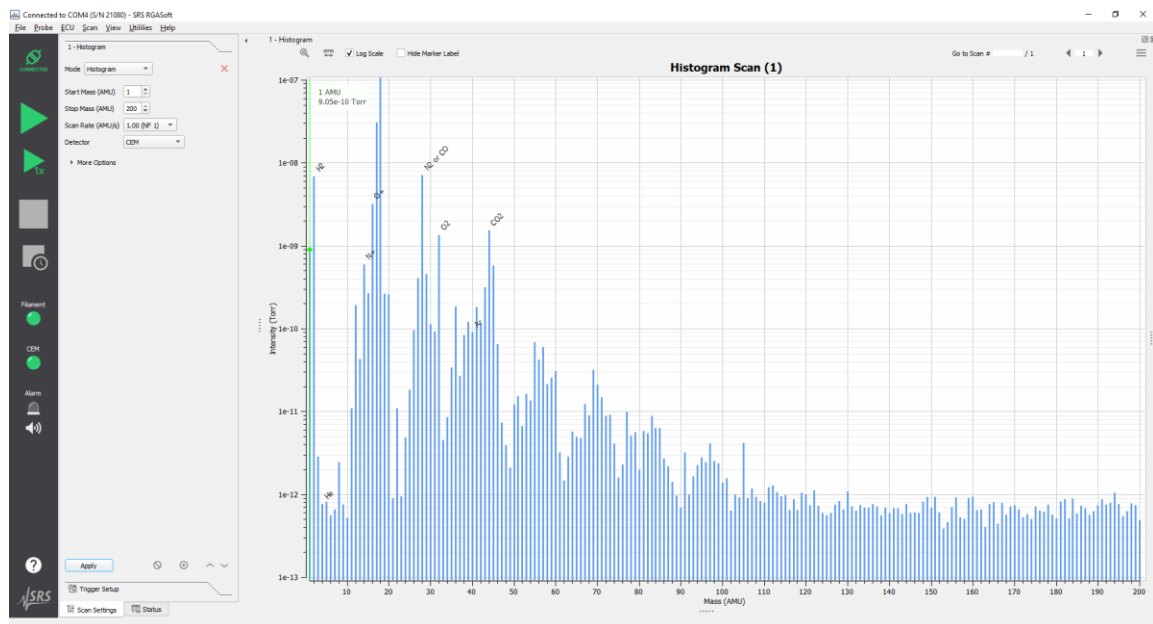
Open the lab with William. We are going to start the RGA and ion source ignition.

08:06 – JSR

Ion source supplies powered on. Ion source pressure jumped from  $1.8\text{E-}8$  to  $9.5\text{E-}7$  torr. Not a bad spike.

08:45 – JSR

Eek! It looks like there's a contamination in the science chamber from the previous tests. That and the pressure is a bit higher than normal since we were only going off of the turbo. We'll probably need to spend time cleaning the chambers and back-filling with nitrogen once we take William's experiment out.

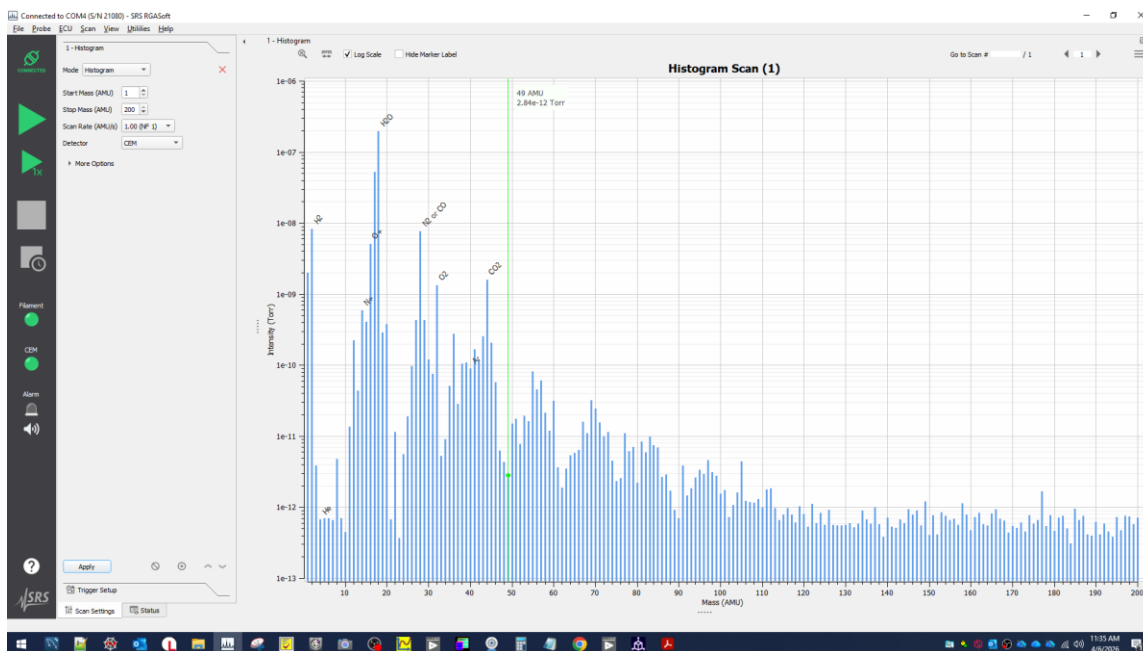


08:51 – JSR

Main Chamber pressure =  $3.7\text{E-}8$  torr. Science Chamber pressure =  $1.9\text{E-}7$  torr. I opened the Main Chamber GV so we could let the East and West Cryos help lower the science chamber pressure.

08:52 – JSR

For reference, see below for the RGA scan of the science chamber after the sapphire ring team installed their project.



08:57 – JSR

William asked an excellent question: “what is sapphire made of?” Turns out it’s a variant of aluminum oxide ( $\text{Al}_2\text{O}_3$ ) which gives it a molar mass of  $2 \times 27 + 3 \times 16 = 102$ . This makes that lone spike there awfully suspicious. Anyhow, good to know for later.

09:02 – JSR

I’m closing the p-leak down to target  $\sim 5\text{E-}7$  torr for the

09:34 – WJL

Lengying opened the science chamber gate valve. Science chamber pressure was  $1.37\text{E-}7$  Torr.

09:42 – WJL

Jamie removed the ABM aft cone SHV-5 cable. This is for the ABM 40V bias plate. We are borrowing it for William Li’s GEMINI experiment. This belongs in the bottom port of the ABM bias plate.

09:51 – LYK

Ready to ramp up MCP to 800 V. The SC pressure is at  $1.2\text{E-}7$  Torr we are going to park the MCP at low voltage to allow outgassing and hopefully allowing the SC pressure to go down.

09:56 – LYK

We are at 800 V for MCP. SC pressure is  $1.2\text{E-}7$  Torr; MC:  $4.3\text{E-}8$  Torr

We will let the MCP sit here for a while.

10:07 – JSR

Pressure is  $1.19\text{E-}7$  torr in the science chamber. Based on the current trends, William and I discerned that it probably won't dip below  $1.0\text{E-}7$  for at least another 3 hours. Therefore, we will proceed, but with caution:

- 1) We will slowly ramp the MCP voltage when the time comes (currently at 800 V and 0.399 mA).
- 2) We will drop the ion source pressure (currently at  $8\text{E-}7$ , target will be  $5\text{E-}7$  torr)
- 3) We will look for any signs of outgassing (little pressure spikes) in the science chamber along the way.

10:13 – JSR

I adjusted the P-Leak down to 34,250 steps which puts the ion source pressure at  $5\text{E-}7$  torr.

10:23 – JSR

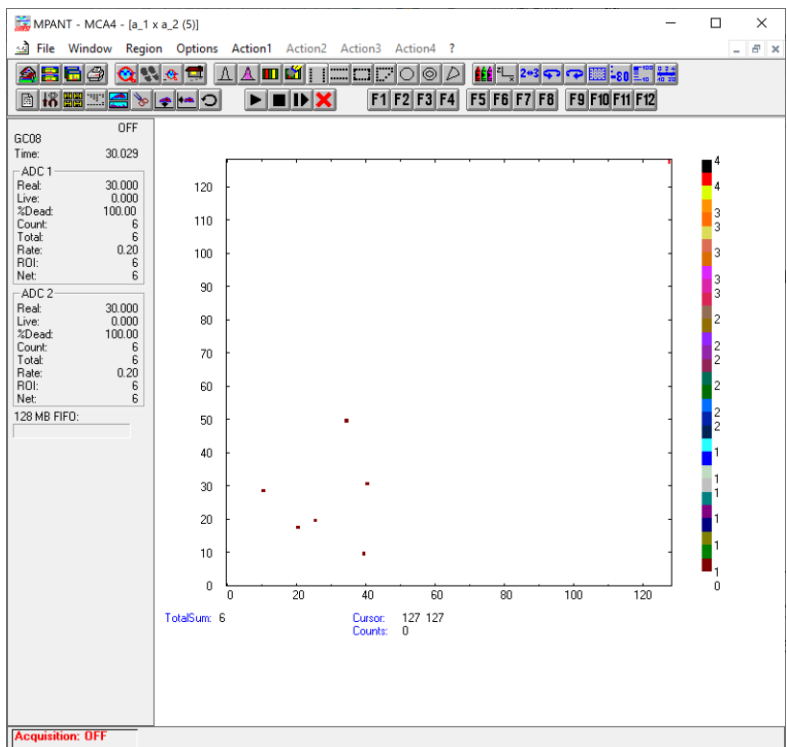
We're at 1100 V and no pressure spikes in the science chamber or dark counts in the MCP, which is as expected (anticipated "dark counts" should start around 1400 V). We still have the beamline GV closed. We will open this after we see dark counts.

10:28 – JSR

We still did not see any counts with 1400 V on the MCP. We've seen no spikes in the science chamber so far...

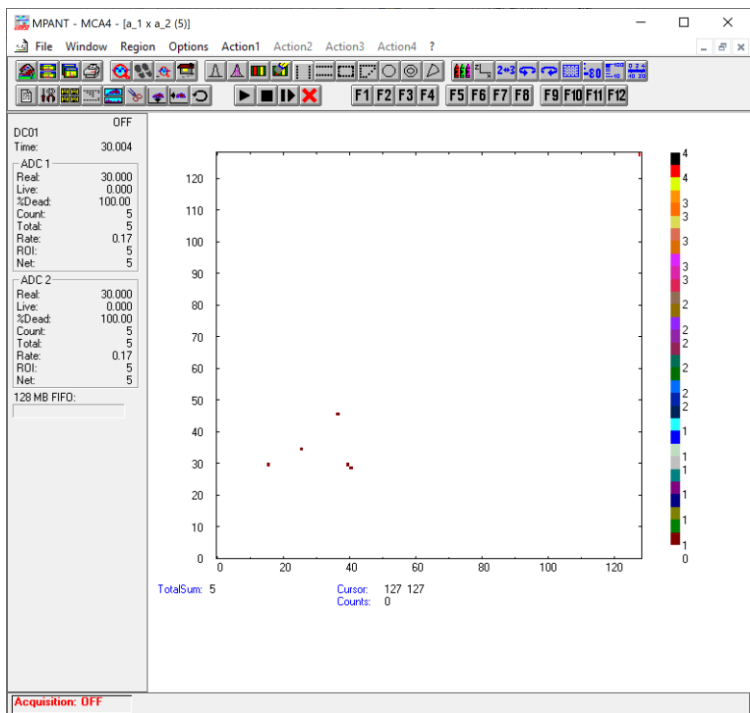
10:32 – JSR

We saw our first counts at 1500 V on the MCP. Current is 0.751 mA. These weren't completely "dark" because the science chamber ion gauge was on. We saw 0.2 Hz (6 counts in 30 sec) and saved this file as "**GC08**". Now we're going to make the true dark counts file and save it as "**DC01**".



10:35 – JSR

The rate with the Science Chamber gauge off is at about the same level: 0.17 Hz, 5 total counts in 30 seconds.



10:37 – JSR

I opened the beamline GV. I think we'll need to turn the science chamber gauge back on because we'll need to make sure we don't have any discharge events throughout these tests (including when William applies voltage to the GEMINI deflector plates). It doesn't seem like the gauge is contributing much (if at all) to the dark counts rate anyway.

10:39 – JSR

We saved file "T01" to capture this data at 1500 V on the MCP, 0.751 mA current, beamline GV open, Ion source Pressure = 4.98E-7 torr, Ion Turbo Pressure is at 1.94E-9 torr.

Total counts were: 2 in 30 seconds, which corresponds to a rate of 0.07 Hz.

10:48 – JSR

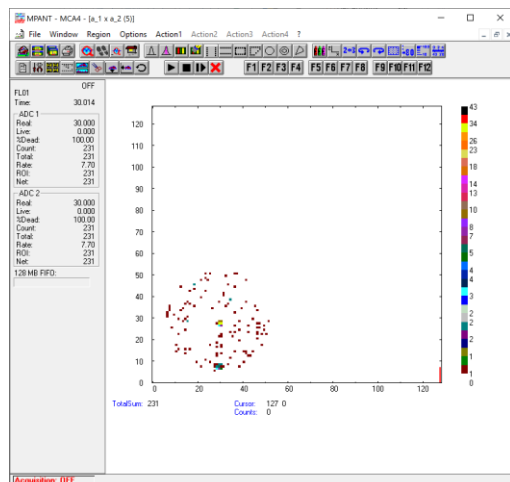
We're proceeding with the ramp up. 1600 V on the MCP with a current of 0.801 mA. We saw 11 total counts (0.37 Hz), so maybe we're getting some of the beam (file **GC09**). Now I'm turning on the science chamber ion gauge. William is also carefully recording this gain curve test in the spreadsheet. At some point, we should see the beam emerge through the pinhole from the GEMINI apparatus. I'll log that "first light" result.

11:11 – JSR

Gain curve tests complete. 1800 V is where we'll target on the MCP today. That gave us a 14.5 Hz rate. The current was similar to previous operations of the MCP.

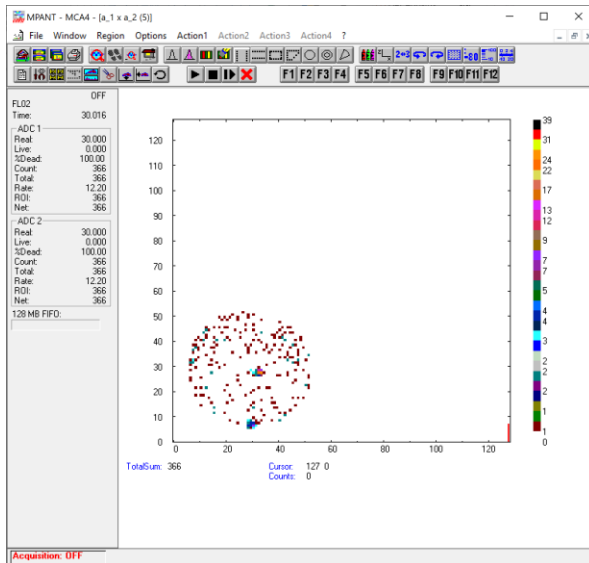
11:15 – JSR

First light! Hooray! Deflector power supplies are powered on, but at their lowest setting (in this case -37 V and +39 V. Currents are 0.000 to 0.001 mA. These data are saved in "FL01".



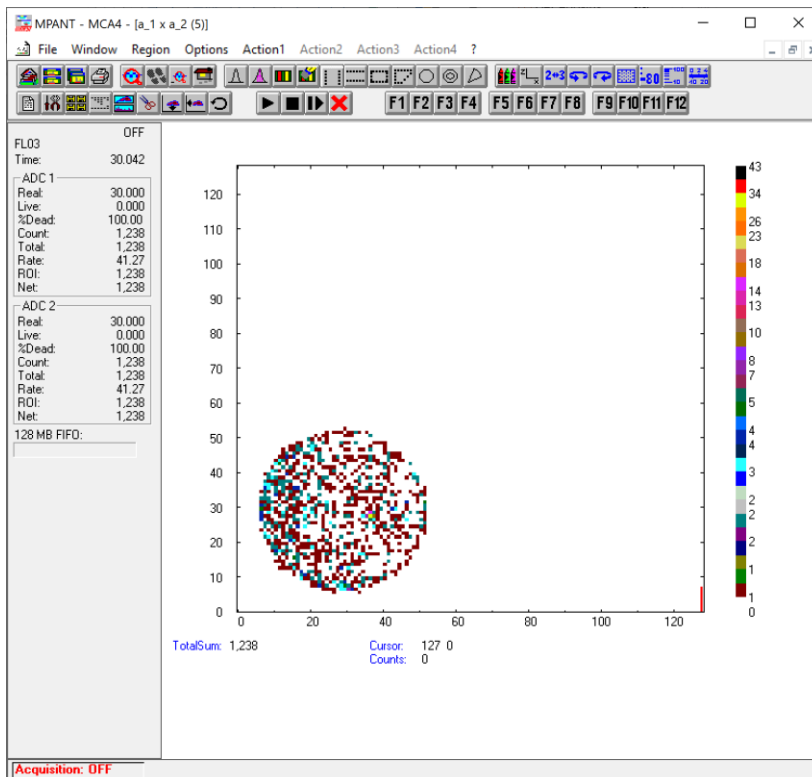
11:23 – JSR

Hooray! At +/-250 V on the deflectors, William steered the proton beam! Looks like it moved ~5 mm.



11:27 – JSR

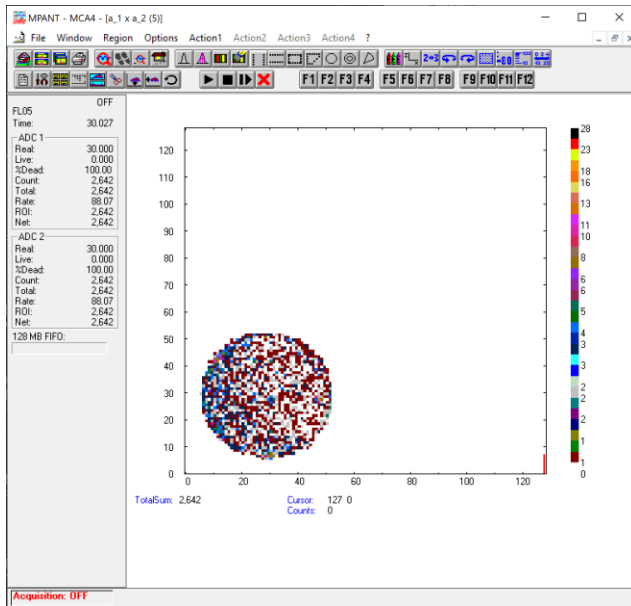
William steered the beam again, with the dot now centered at around 38 mm, but there's A LOT more scattering, so the actually pinpoint is hard to see.



11:34 – JSR

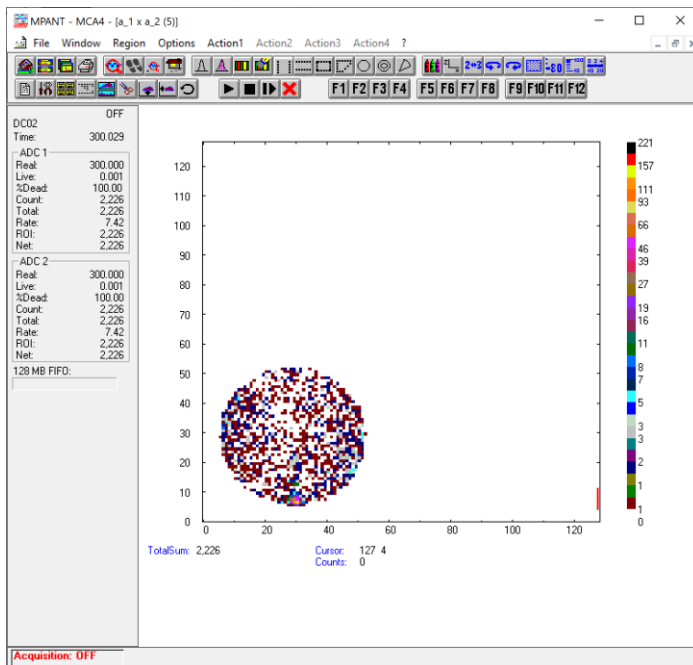
At William's target deflector voltages ( $\pm 1083$  V), the proton beam has moved to the edge of the MCP, exactly as William intended. Hooray! There is a lot of scattering. Some of it could

be due to the neutral beam contamination. But right now we're going through the open aperture. Maybe a chunk of this will be eliminated



12:11 – JSR

We took “dark counts” with the beamline GV closed but science chamber ion gauge on, so maybe that odd “structure” we see (and the scattering) can be explained by the ion gauge! Ops. That would explain a lot. We saved this in file DC02.



12:15 – JSR

I turned off the science chamber ion gauge. Starting “DC03” which should reflect the real dark counts now.

12:20 – JSR

I tweaked the P-leak to 37,000 steps.

12:26 – JSR

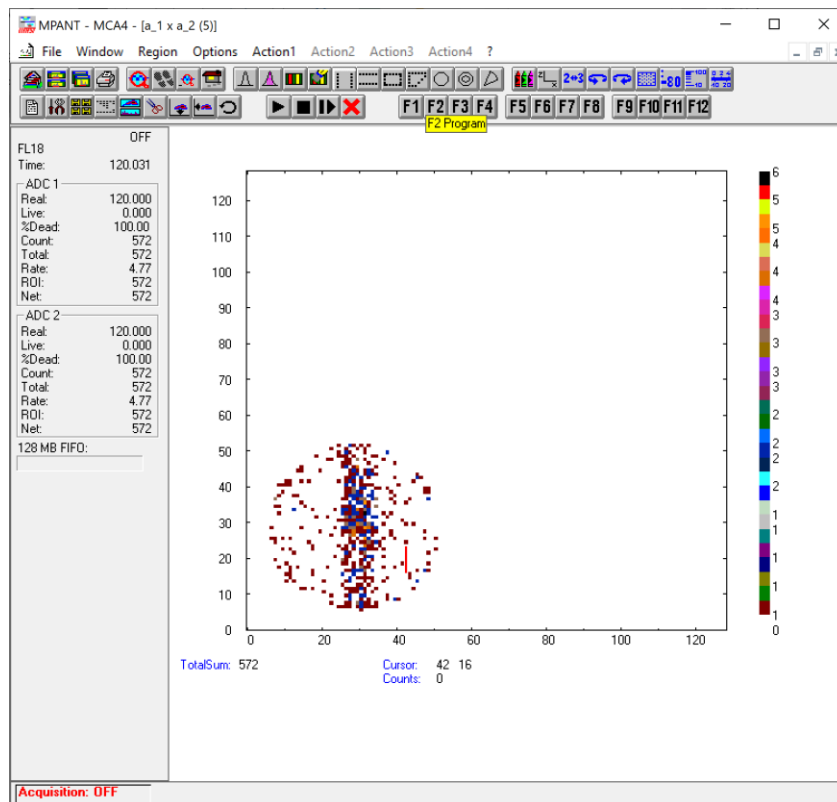
We re-opened the beamline GV. IS =  $1.0\text{E-}6$  torr. IT =  $3.6\text{E-}9$  torr. Deflectors at  $\pm 1084$  V. The stage is rotated to amorphous carbon (54.12 deg). This might be the actual real first light!

13:03 – LYK

Close the beamline GV twice. Once to change the IS pressure to  $5\text{E-}6$  Torr. Now at IS =  $1\text{E-}5$  Torr.

13:13 – LYK

At  $2\text{E-}5$  Torr, we saw about 5 Hz. That means with 120 second dwell, the uncertainty is  $\sim 5\%$ .



We set the deflection plate to 0 V and ramp MCP down to 1400 V. Turn on the SC ion gauge to monitor the status.

13:20 – LYK

Students are still in the lab. Will ask Ted to cover me while I have a lunch break.

15:09 – JSR

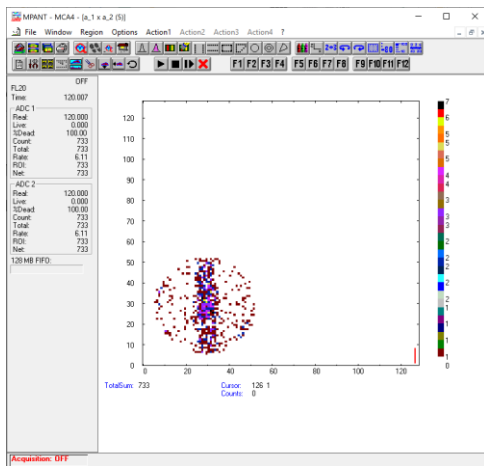
Back to William's GEMINI measurements. I turned off the gauge and ramped up the Quantar MCP back to 1800 V (current = 0.902 mA). William set his deflector voltages to +/-750 volts. The stage is currently on the amorphous carbon.

15:15 – JSR

I opened the beamline GV. Let the testing resume!

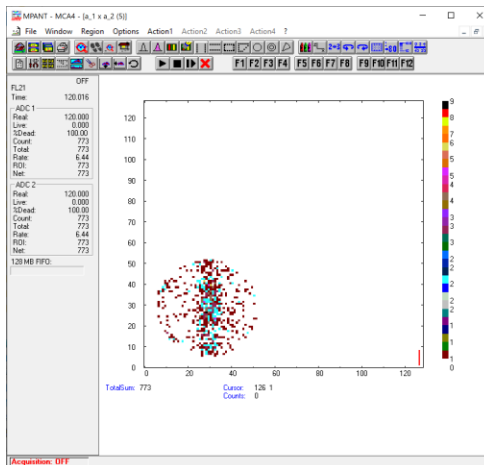
15:18 – JSR

Here's the result with deflectors at +/-750 volts and Ion source pressure at 1.7E-5 torr.



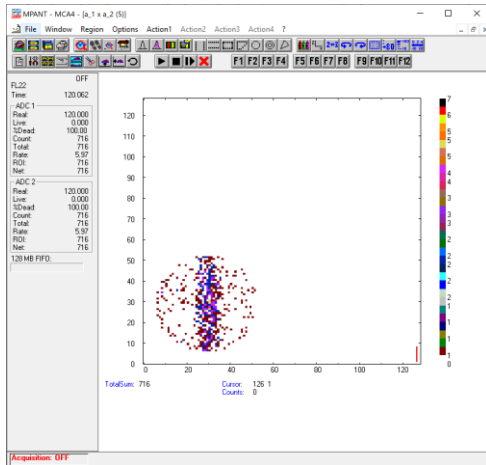
15:26 – JSR

Here's the result with deflectors at +/-500 volts and Ion source pressure at 1.7E-5 torr.



15:29 – JSR

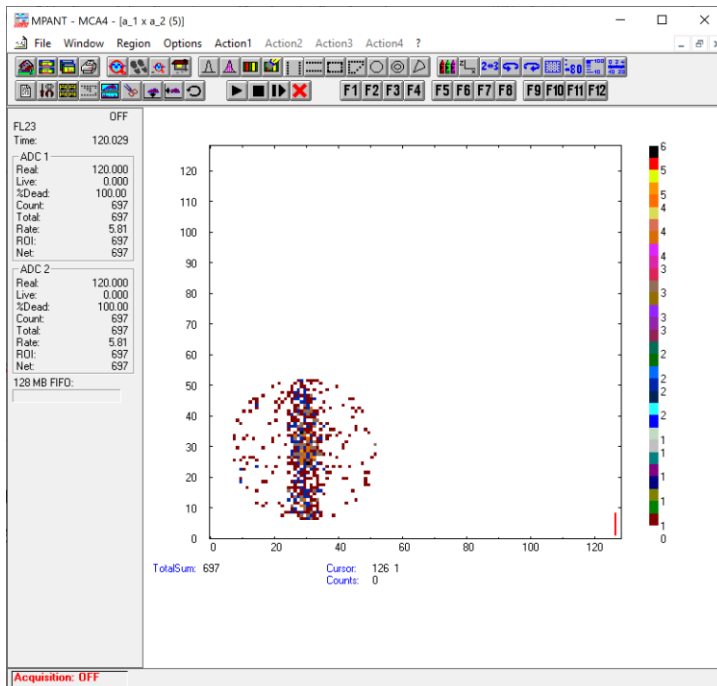
Here's the result with deflectors at +/-250 volts and Ion source pressure at 1.7E-5 torr (FL22).



15:35 – JSR

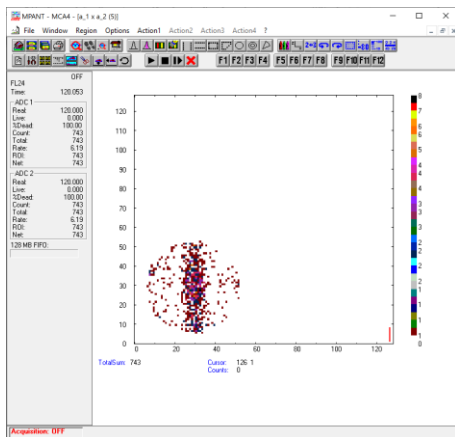
Here's the result with deflectors at +/-0 volts (in practice, +39 and -28 V) and Ion source pressure at 1.7E-5 torr (FL23). Hmm. I mean, maybe the “hot spot” feature is different?

But the distribution doesn't seem to have moved.



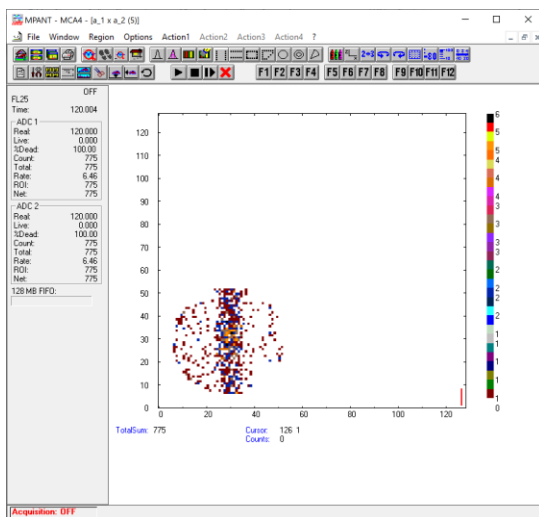
15:40 – JSR

Here's the result with deflectors at +/-1250 volts and Ion source pressure at 1.7E-5 torr (FL24).



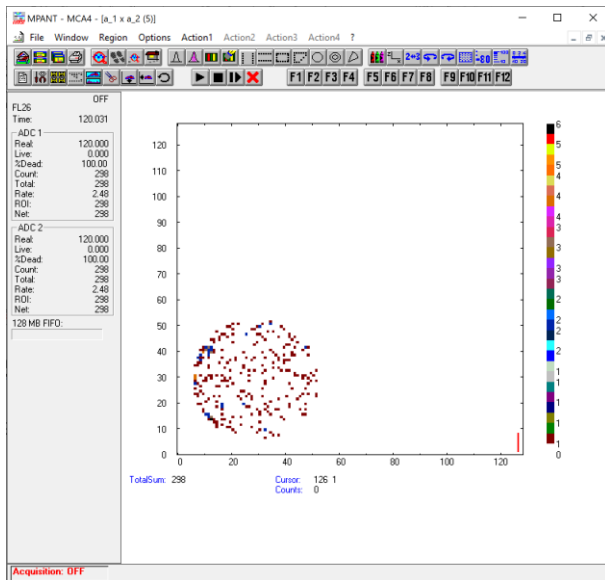
15:45 – JSR

Here's the result with deflectors at +/-2000 volts and Ion source pressure at 1.7E-5 torr (FL25). This should have completely deflected the negative and positive charges away...so have we just been seeing neutrals the whole time?



15:49 – JSR

Same settings, but with HC off.



So the ion beam is the incident cause of the stripe, but we don't quite understand what's happening on the back end...

15:50 – JSR

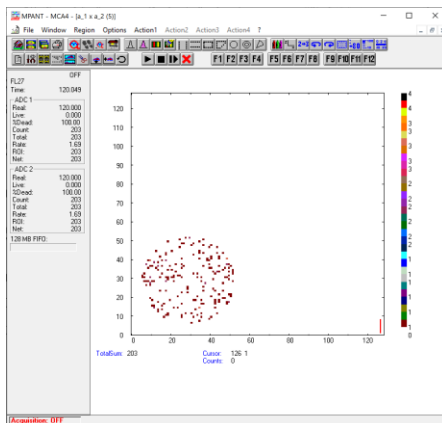
I closed the beamline GV. We're going to rotate on to the graphene foil.

15:55 – JSR

I opened the beamline GV and we started measurements on the graphene foil with the ion source pressure at  $1.66\text{E-}5$  torr and ion turbo pressure at  $5.63\text{E-}8$  torr. Oops – had to also turn the HC back on too. 😊

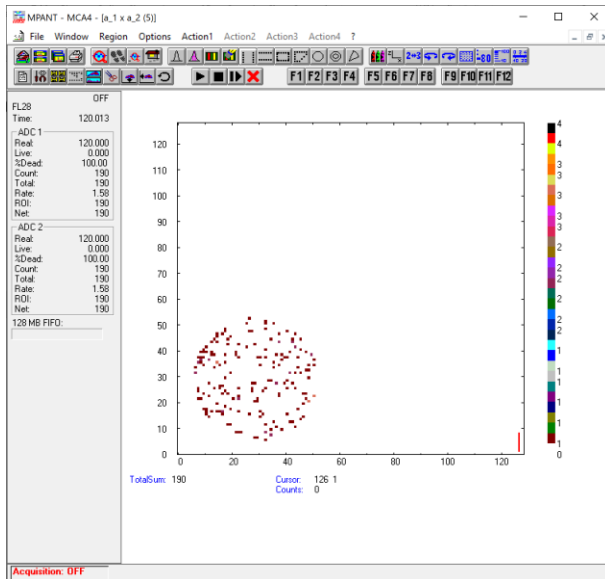
16:00 – JSR

Well...here's our very first measurements with a graphene foil! The results are quite...puzzling! Plates are set at  $\pm 1084$  V. Nothing else has changed other than the foil.



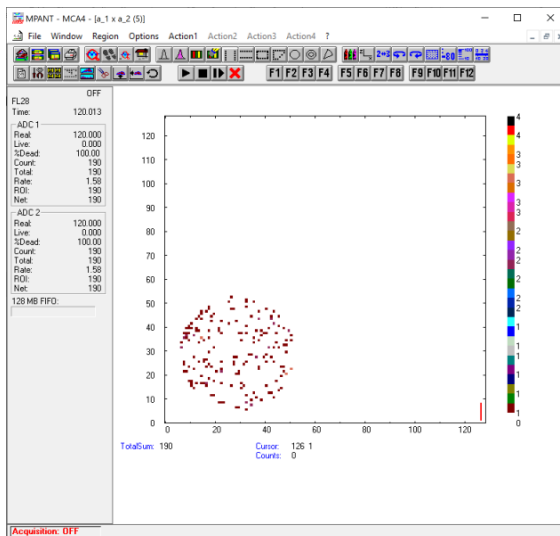
16:01 – JSR

No obvious change with the plates set to  $\pm 160$  V.



16:09 – JSR

Wow. With 0 dialed in on the plates, we get something close to dark counts. Could the graphene be blocking the 1 keV particles nearly completely?

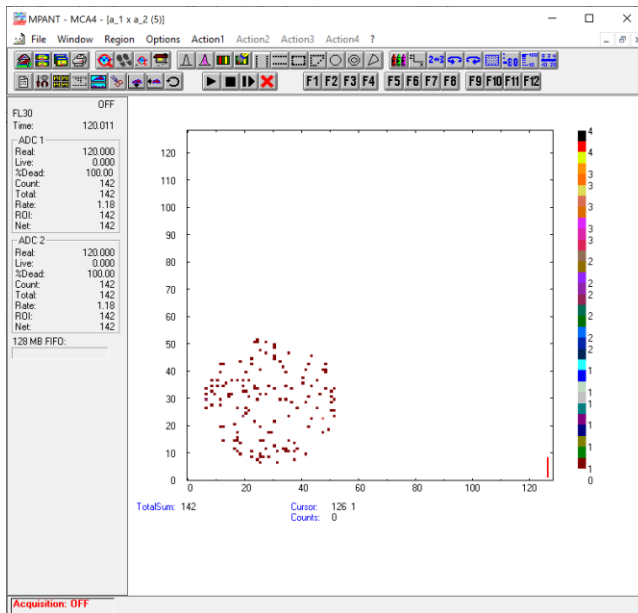


16:14 -- JSR

We closed the beamline GV and dialed in 4 keV protons. We won't be using any deflector voltages here, but this will tell us if we can get any part of the beam through the graphene foil at all!

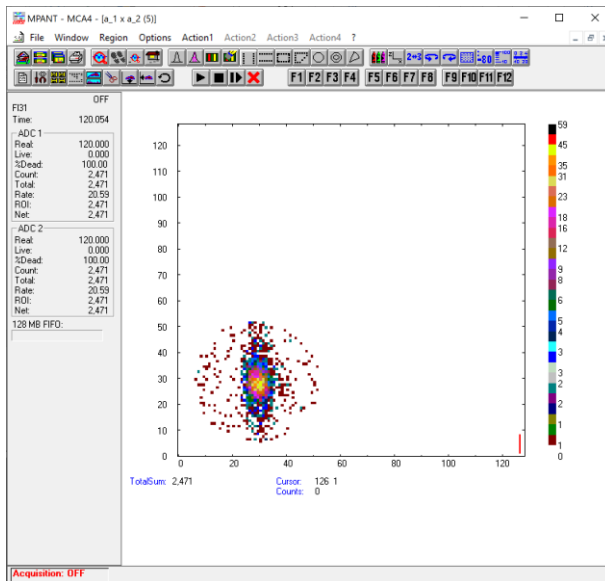
16:18 – JSR

No beam through the graphene!



16:25 – JSR

We were so puzzled that we closed the beamline GV, left everything else untouched and only rotated the amorphous carbon foil into the beam. Here's what we got from the acquired data. There's obviously a beam going through AND the scattering ellipse is smaller than the diameter of the MCP, which is what theory would predict.

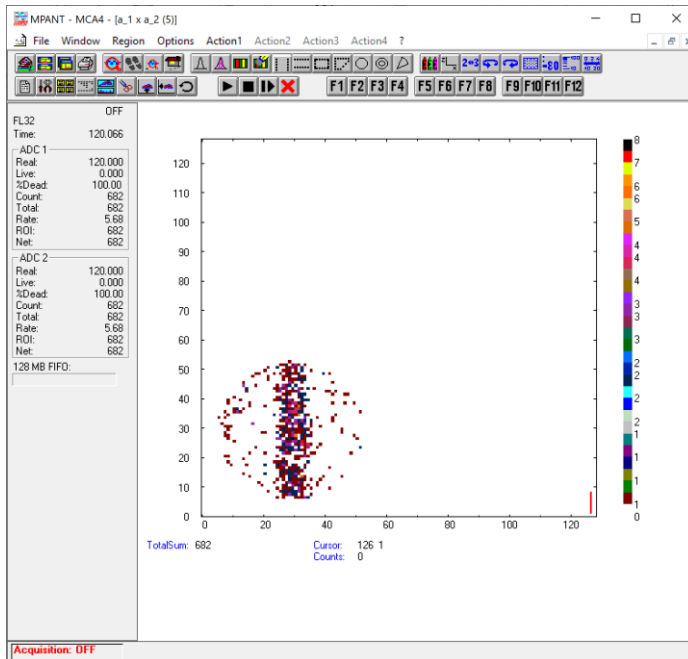


16:29 – JSR

I changed the beam energy back down to target 1 keV protons.

16:33 – JSR

We kept the deflectors at 0 V (really -29 and +39 V) and saw the usual stripe come back (FL32):



The rate is consistent with what we previously measured in this configuration (5.81 Hz; FL23).

16:34 – JSR

Lastly, per William's request, we're going to drop the pressures (currently IT =  $1.6 \times 10^{-5}$  torr, IS =  $5.6 \times 10^{-8}$  torr) and get back to some safe low value to confirm that the deflectors do, in fact, steer the pinhole beam (no foil configuration).

16:38 – JSR

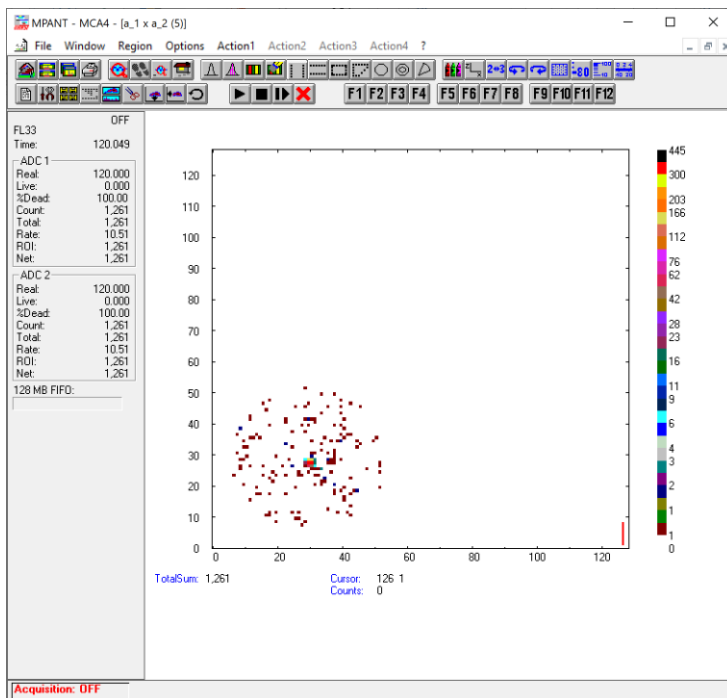
I brought the ion source pressure down to  $1.1 \times 10^{-6}$  torr. I'll put it a little below that (targeting  $7.5 \times 10^{-7}$  torr), but a little higher than this morning ( $5 \times 10^{-7}$  torr).

16:41 – JSR

P-Leak now at 34,750 steps. Ion source pressure between  $7.3$  and  $7.5 \times 10^{-7}$  torr.

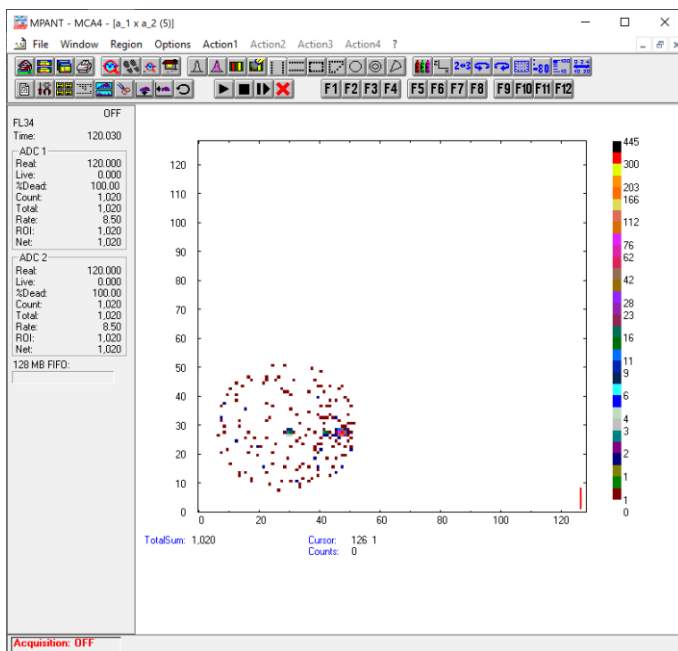
16:42 – JSR

William opened the beamline GV. FL33 captures the dot (same as this morning) with 0 V on the deflectors.



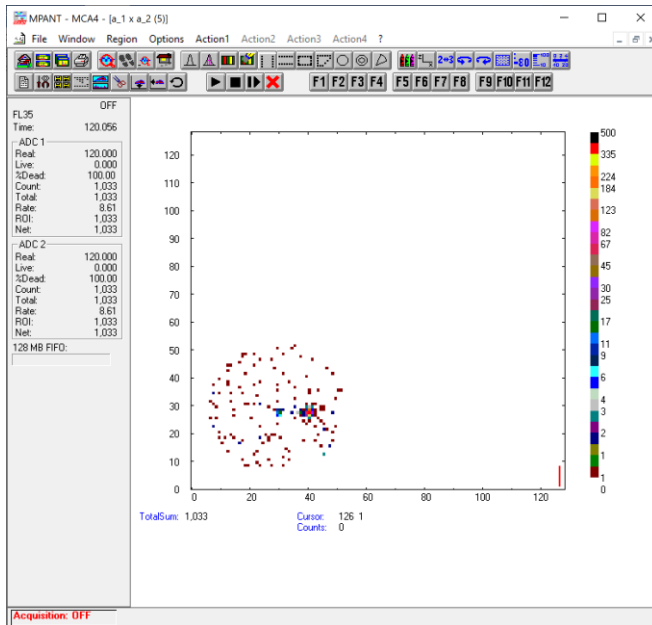
16:42 – JSR

FL34 captures the beam dot location with  $\pm 1084$  V on the deflectors. So the deflectors are clearly moving the beam! Maybe in this case there's some scattering at the edges. And there does appear to be a neutral component coming through as well.



16:48 – JSR

Okay! One final deflector setting (+/-750 V; FL35) through the pinhole (no foils) and this indeed confirms we are properly steering the beam (without a foil, anyway).



17:02 – WJL

Lengy turned off the high voltage power supply on the ion source. We turned the science chamber ion gauge back on. The science chamber pressure is 2.2E-7 Torr.

17:41 – JSR

Ion Source, Deflector plates, HC, MCP power supplies are all powered down. We're leaving the Main Chamber GV open to help bring the science chamber to lower pressures.

17:43 – JSR

Time to close the lab.